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# Using a service blueprint to alleviate barriers and challenges of digital public service adoption

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## Abstract

Digital transformation has been the latest uptake in service development, and many private and public services are being moved from physical services to digital ones. At the same time, new services are being developed directly as digital services. Citizens can often decide whether to adopt the service in the private sector. Still, in the public sector, there is an inherent mandate that the service needs to be adopted, or the citizen may lack a vital service in their life. This means that public service designers need to consider the needs and requirements of different citizens when developing or updating public services, especially in a digital format. Citizens' different demographics and backgrounds will affect how they accept new services and whether they will adopt them. This paper aims to explore the challenges and barriers in digital public service adoption and propose the use of a service blueprint to provide a service design and development template for public administrations to alleviate the challenges of acceptance and adoption. The service blueprint will incorporate best practices and recommendations gathered from literature and end-users through surveys and interviews in the H2020 IMPULSE project.

**Keywords:** service blueprint, digital service, public service, service design, service adoption, user acceptance.

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## 1 Introduction

With the rise of smartphones and people constantly being online, many services are offered in a digital format. New services are created with the latest generation of people in mind and provided in digital form. At the same time, old services are transformed into the digital landscape. Most services offered by public entities are slowly following in their footsteps by catering more towards the digital side and reducing the physical services to reduce costs and required time. However, the adoption of digital services (new or old) has its challenges and barriers that stem from users, technology, and culture (Mu & Wang, 2022; Wilson & Mergel, 2022). Many needs and requirements must be considered during the design and development of a service to improve the adoption rate, and keeping track of everything can be demanding.

This paper focuses on utilising a service blueprint technique (Shostack, 1984) to help service providers consider and alleviate the barriers and challenges of digital public service adoption. The focus is on public services rather than private ones because they have to be offered to a broader audience, while private services can limit their user base as needed. This paper proposes a service blueprint template containing recommendations and best practices to be considered while designing, implementing and adopting the service to better address the barriers in acceptance and adoption.

The best practices and recommendations are gathered from the results of various activities in the H2020 Identity Management in PUbLic Services (IMPULSE)[1] project, such as end-user surveys, and interviews with public administrations and service providers. The contribution of this paper is to provide guidance for future public service developers on the challenges of acceptance and adoption, and possible methods for alleviating the encountered barriers through the proposed service blueprint model.

## 2 Related work

One key component of adopting a new technology or service is *user acceptance*. User acceptance significantly affects people's willingness to use the technology or service. With the Technology Acceptance Model (TAM) (Davis, 1989; Venkatesh & Bala, 2008), it is possible to map different user acceptance variables. The original model defines four variables that affect the usefulness and ease of use, which lead to acceptance: *individual differences, system characteristics, social influence and facilitating conditions*. These four variables are generic and can be adapted to a more specific context. For example, Guggenberger et al. (2023) modified and adopted the TAM to the self-sovereign identity context by defining eight variables. These variables include *regulation and operation, privacy and data protection, output quality, result demonstration, support, educational work, number of use cases,* and *user experience.* In addition, the input variables also affect trust in the technology, ease of use and usefulness. All eight variables can be mapped into the four original variables, but they give a more specific and detailed context to using TAM.

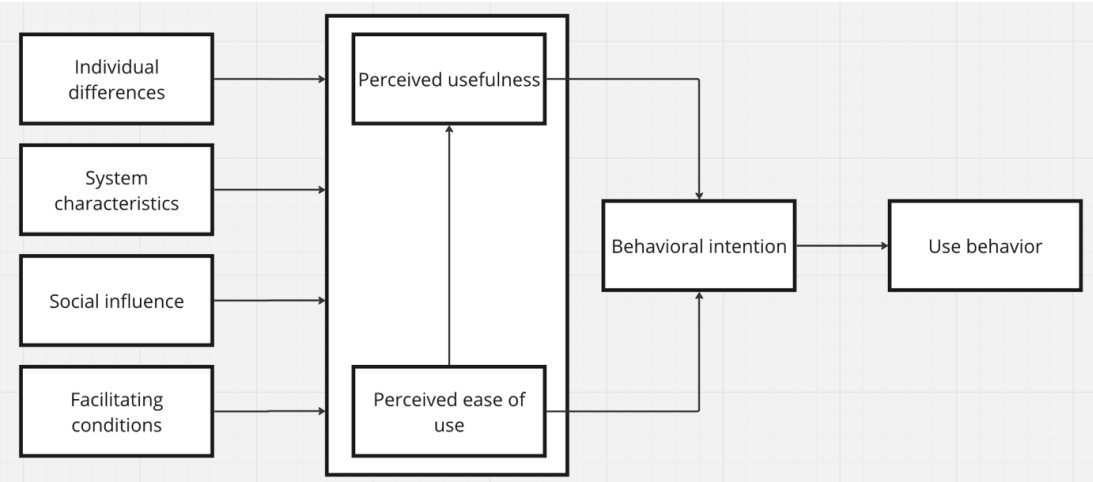



**Figure 1.** Technology acceptance model (Venkatesh & Bala, 2008).

When examining public services with TAM in mind, a variety of variables can be noticed. Public services are offered to various groups of people with widely different demographics, such as different ages, educational backgrounds, cultures, skills and competencies, and many others. These *individual differences significantly impact* the acceptance and adoption of digital services. In addition, the characteristics of the service, such as privacy and data protection, regulation and operation, and quality of the service, contribute to the adoption rate. These design decisions from

1. https://www.impulse-h2020.eu


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the implementation and development need to be considered. The *social influence* will be reflected in the *number of use cases* (if a similar service is used elsewhere). Finally, the *support* and *user experience* are facilitating conditions in the TAM.

When aiming to increase adoption and acceptance, it is crucial to involve end-users to consider individual differences in the best way. Involving end-users will bring insights into how the service design and implementation will affect them. Within the context of public services, this means communicating and engaging with the public either directly or indirectly. It is imperative to consider the public's needs and requirements and appropriately align the services for their users.

**Public engagement or public participation** refers to how the public can be involved in design and development. The public can be engaged by directly having them participate in the processes and decision-making, by utilizing open forums or workshops, or ensuring that the public has representatives. There are different levels of engagement, such as collaboration, communication, and consultation. These different levels reflect how much influence the public can have in decision-making. At the highest level, public engagement can be considered a form of co-creation, involving the end-users directly in the design and development (Rowe & Frewer, 2005; Evans, 2020).

In the case of public service design and development, including the public in the process is vital. Without understanding the needs and requirements of the users, the service may be subpar and not the be adopted or cause irritation and negative emotions within the public.

## 3 Data collection

Data for this paper has been collected within the IMPULSE project. In the project, six different public administrations from five different countries piloted a new electronic identity (eID) solution for signing in to their public service. The new eID solution was designed, developed and integrated over the course of the project. After the eID solution had been created, the public administrations recruited citizens for two rounds of pilot testing.

During both test rounds, the participants were asked to answer end-user surveys and participate in a solo or group interview to provide feedback about the new eID solution and about their experiences with digital services in general. More than a hundred stakeholders were involved in providing their feedback during the project.

After the pilot testing, the answers were shared and discussed with the public administrations to collect their thoughts and views on digital public service development from the point of view of public administration. Based on the data from the citizens as well as the discussions with the public administrations, different barriers and challenges were identified.

## 4 Challenges and barriers

Different factors bring barriers and challenges to implementing public digital services which relate to end-users as well as public administrators or policymakers (Mu & Wang, 2022; Wilson & Mergel, 2022). The different factors can be tied to the four variables mentioned in the TAM.

Table 1 shows the barriers and challenges based on the data from the different stakeholders within the IMPULSE project. These barriers and challenges are described in more detail later based on the data from the project and supported by various literature.


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**Table 1.** Barriers and challenges.

| Barrier / Challenge | Explanation | TAM variable |
|---|---|---|
| Age difference | Age difference amongst users may affect how they accept and adopt changes | Individual differences |
| Level of knowledge and education | Education may have an effect on the users' knowledge and willingness to try new things | Individual differences |
| Technical competency and available technology | Users will have different levels of competency and comfort on how to use digital technologies. Additionally, not all users may have the required technology readily available. | Individual differences |
| Privacy concerns | Users have varying levels of privacy concerns, making some more willing to share personal information | System & individual characteristics |
| Security concerns | Users will have security concerns that need to be properly addressed and explained | System & individual characteristics |
| Value vs. Required work | The service needs to bring value to the users and it should be higher than the required work to achieve it | System characteristics |
| Previous experience | Previous experiences will have a positive or negative effect on how users perceive similar services | Social influence |
| Standards and regulations | Different countries have different legal and ethical standards and regulations that need to be followed | Facilitating conditions |

**Age difference:** The user group of public services is heterogeneous, with ages ranging from underage to elderly. A user's age will affect their competence in utilising different technologies. For example, many older people may struggle with new technologies and have internal barriers against sudden change. If a service has been provided in one way for many years and suddenly changed without any grace period, the users are more than likely to reject the changes. On the other hand, younger users may be more adept with new digital services and technologies. They would adopt them faster, especially if they have grown up in a digital environment. The policymakers should consider all these different people and how to handle their needs and requirements. One suitable method is first to encourage the younger generation to adopt the digital service and slowly have the older generation shift from the previous service into the new one. The change should not be immediate but rather slowly have people accustomed to the new service before phasing out the old one to avoid problems accessing the provided service (Friedhoff et al., 2023).

**Level of knowledge and education:** Within the same age group, users will have varied levels of education and knowledge. While education does not necessarily correlate to knowledge, as users can be self-educated, there will be a difference in the level of general knowledge based on education levels. This means some users require less help and support when using digital services. Most users would want to know, at least on a basic level, how the technologies behind a digital service work. More knowledgeable users will need less explanation to convine them to trust and use the digital service. Policymakers should inform users about how the digital service works while considering the varying levels of knowledge the users have (Cubric, 2020; Friedhoff et al., 2023; Guggenberger et al., 2023).

**Technical competency and available technology**: The users will have different levels of technical competencies and available technologies they can utilise (Keil et al., 2022). While most people have a smartphone, some still do not possess one. In addition, fewer people have tablets, laptops, computers, or access to such devices. This means there will be people who rely on


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physical services or require the necessary devices to be arranged by the service provider. When a service is only provided digitally, there will be complaints from those users who cannot access it because of the lack of technology. With public services, it is necessary to ensure that all users have an equal opportunity to access the service. Even if the users have the required device, the digital service might not work correctly for various reasons, such as the given operating system not being supported or the lack of specific features on the device. All of these need to be considered by policymakers to ensure that end users are not left without service.

**Security and privacy concerns**: While everyone has privacy and security concerns, the comfort level differs between individuals. For example, some people are more willing to give access and share their personal information with a reliable and credible service provider to be able to use a service. In contrast, others carefully read the terms and conditions before accepting anything. Privacy concerns are often related to security concerns, as they lead users to doubt the safety and privacy of the service (Keil et al., 2022). If security concerns are based on the lack of knowledge of the service or technology being used, the concerns can be resolved by explaining the service and technology to the users. While terms and conditions provide information regarding privacy matters and data sharing, they are a poor way to share this knowledge as few people have the time and energy to go through heavy legal texts. It would be better to provide users with simplified and easy-to-digest information about security and privacy aspects (Andersen, 2021; Guggenberger et al., 2023). These individual levels of concerns are also affected on the system level of concerns and guarantees, how a system guarantees privacy and security and what issues there have been in the past. How a system provides security and handles privacy varies across domains and use-cases based on legislation as well as common operating models. Some systems, such as healthcare and banking, have very strict rules and standards they need to adhere to while other systems, such as social media, have less rules to follow. These system characteristics affect the initial concerns while the individual characteristics enhance these concerns based on their individual comfort level.

**Value vs. required work**: To have more people use a service, the service needs to bring value to the user. The value can be tangible, such as money, or intangible, such as enjoyment or practicality. If multiple providers offer the same service, users will select the one that brings them the most value with the least required work. The required work can refer to software installation, account creation, ease of use, etc. The provided digital service should be as convenient as possible and provide the most value for users. If the service is too difficult for users, they will stop using it unless necessary (such as electronic health records) (Andersen, 2021; Portz et al., 2019; Sohn & Kwon, 2020).

**Previous experiences**: Previous experiences can have a drastic impact, positive or negative, on how users will accept a digital service and technology. If users have prior bad experiences, they will have more doubt and negative feelings towards the new service even if there are no problems with it. Similarly, if users have good experiences, they are more accepting and willing to give a chance to the digital service. If users have no prior experience, that can act as a barrier, as there might be confusion, discomfort and doubt initially. However, after getting accustomed to something new over time, the users might be more positive towards the service. These previous experiences should be considered by policymakers when designing and developing the service (Cubric, 2020; Friedhoff et al., 2023; Guggenberger et al., 2023; Portz et al., 2019).

**Standards and regulations**: Different countries have their own ethical and legal standards and regulations that will affect what the service can and cannot do or how it should operate. Different regulations should be followed depending on what kind of data is being handled in the service, such as GDPR, medical records, banking, and so on. Following standards in the design and development of a service is a positive aspect overall, but implementing those standards can be


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difficult. Similarly, following legal regulations can be easy if there is no need to consider multiple different countries. For example, there are regulations and guidelines at the European Union level, but each member country may have a specific implementation of how these are filled. So, certain aspects must be considered if a public service deals with international users. If a digital service is being adopted from another country and modified, there can also be difficulties (Custers et al., 2017).

## 5 Service blueprint for digital public service

A service blueprint is a service design and development technique that provides an overview of users' steps and actions and what happens in the background. The service blueprint is constructed from the customer's perspective. A typical service blueprint consists of five elements, as shown in Figure 2:

- *Customer actions:* The steps a customer (end-user) takes during the service delivery.
- *Front stage:* The visible part that the customer interacts with. When using a digital service, this would be the front-end of the service.
- *Backstage:* The invisible part that provides the actions for the front stage. When using a digital service, this would be the backend of the service
- *Support processes:* The supporting processes for the delivery of the service.
- *Physical evidence*: The concrete elements related to each step that influence the customer's perception of the service.

| Physical Evidence | Step 1 | Step 2 | Step 3 | Step 4 | Step 5 | Step 6 | Step 7 | Step 8 |
|---|---|---|---|---|---|---|---|---|
| Customer Actions | | | | | | | | |
| Front of Stage Interactions | | | | | | | | |
| Back of Stage Interactions | | | | | | | | |
| Support Processes | | | | | | | | |

**Figure 2.** Generic version of service blueprint.

It is not always necessary to fill all parts fully, as that depends on what parts of the blueprint the service provider wants to focus on. The service blueprint can be used for different purposes, such as representing the service, diagnosing problems, and designing and developing services (Bitner et al., 2008).

This paper uses the service blueprint for service design and development. The idea is to propose a simplified blueprint for a generic digital public service design that can be followed and filled with additional details based on the service provided. While the traditional service blueprint only considers the service process and delivery, the blueprint presented in this paper will also


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consider the phases before the service delivery, i.e. the design and development phases, as those are also vital and having end-user input is beneficial. The blueprint is designed to be as generic as possible so it can be adapted to various digital service deliveries. While this helps with generalizing the model, it will also act as a limitation as often the service design and development for a public administration is not as simple and will have more complexities and actors.

## 5.1 Blueprint for digital service design and development

Figure 3 presents the complete service blueprint for digital public services, and Figures 4 and 5 present the blueprint in two parts, before and after final implementation. Compared to the traditional service blueprint, it includes steps before the actual service delivery and additional elements in the form of recommendations and best practices. Recommendations and best practices are the main elements in the design and development of a service, as these are things to be considered when creating or modifying public services. As the service blueprint is for a digital service, the customer interacts with an application or a platform (front stage) in a digital environment without any physical interaction during the service usage -phase.

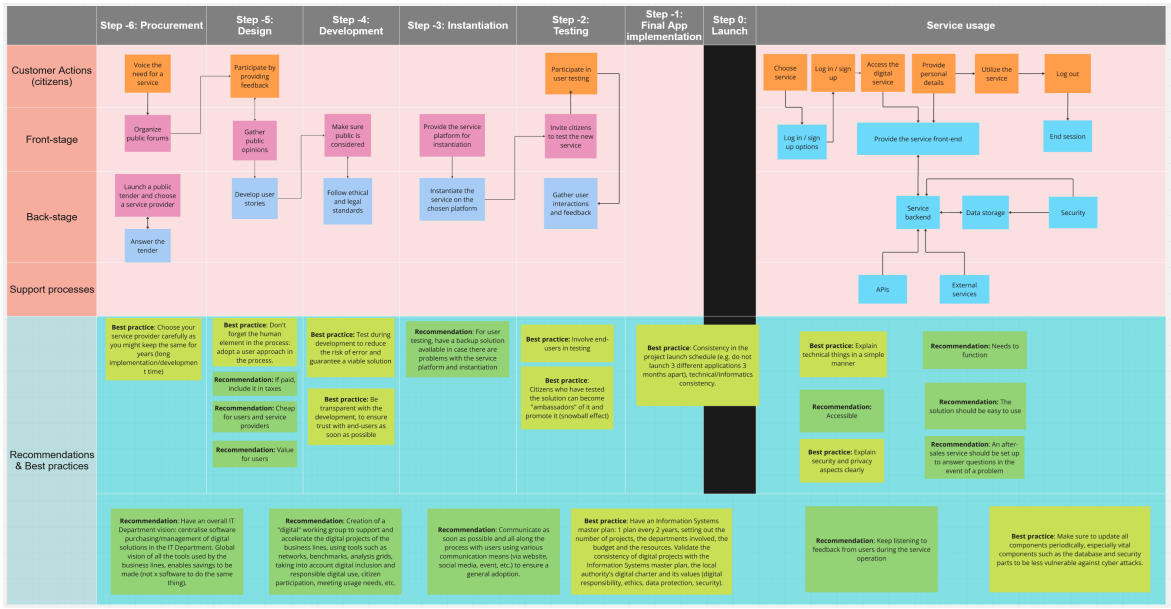



**Figure 3.** Generic digital public service blueprint.

Figure 3 shows the different actions and events with different colours as follows:

- **Red**: Citizen actions that they can take during the service design and development as the target end-users. These actions can also be taken by intermediaries, such as interest groups, political parties, non-governmental organizations, institutions etc. who would then represent the citizens' voices when communicating with the public administration and act as a link or a representative between the public administrations and citizens.
- **Purple**: Public administration actions that the public administrations as service owners can take.
- **Blue**: Technical actions taken by either the technical team or the system.
- **Dark green**: Recommendations for the service design and development.
- **Light green**: Best practices to be followed during the service design and development.


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As the blueprint shows, citizen actions are taken in the customer lane, while public administration actions are primarily in the front stage (before launch). The technical actions are taken in the backstage by the technical team before the launch and by the system during service usage.

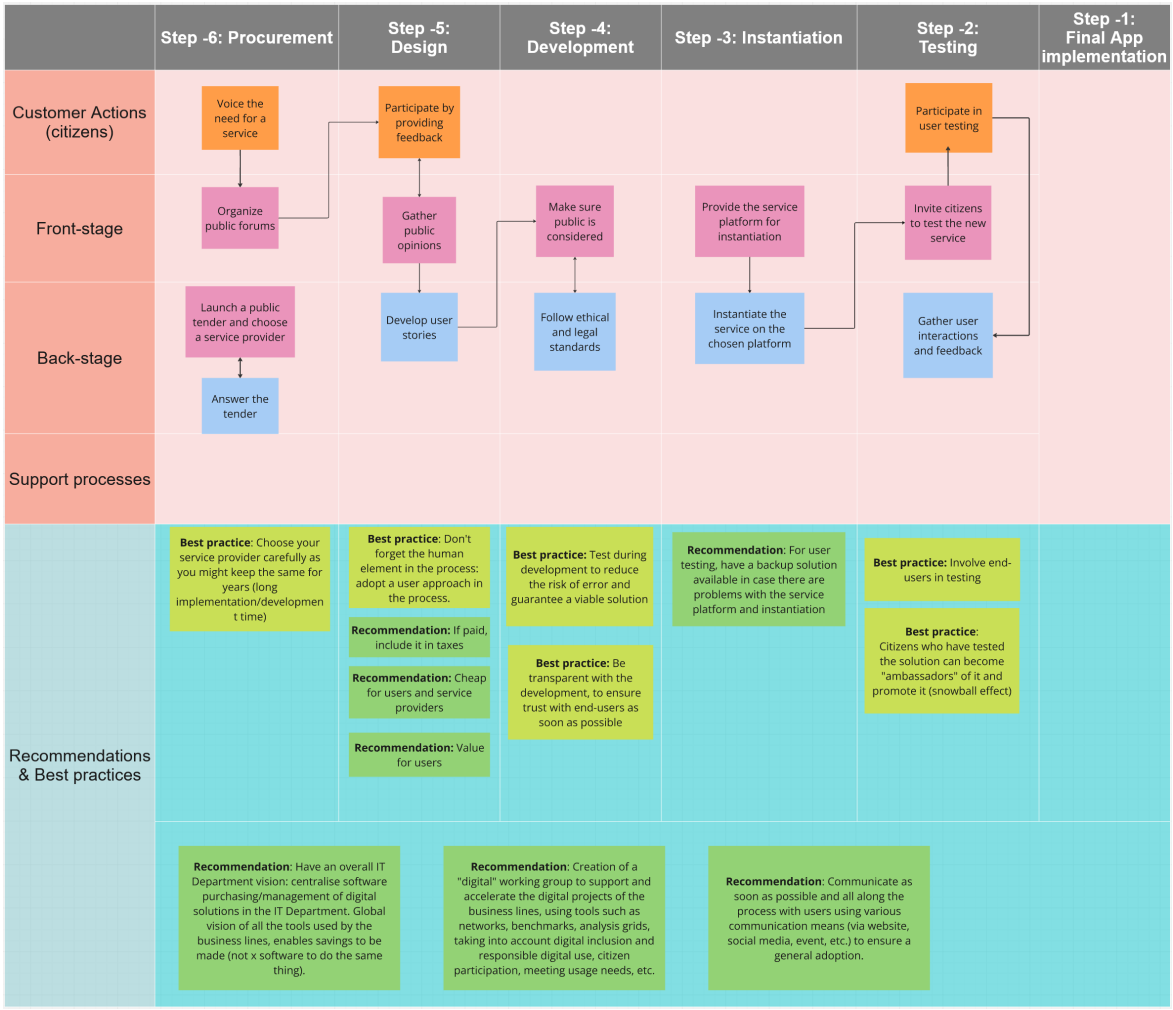



**Figure 4.** Service blueprint, part 1 before final implementation.

Figure 4 presents the first half of the service blueprint, which consists of the phases before the final implementation and launch of the digital service. The whole service design and development starts with the procurement phase and goes through design, development and testing before finalising the implementation and launching. Each phase has its own best practices and recommendations, and there are some recommendations and best practices for the whole service development process, as shown in the last row of the blueprint. In addition, the barriers and challenges presented in Table 1 will also be discussed within the different phases, how they may manifest themselves, and how they could possibly be taken into consideration. Some steps might only be relevant for some digital public service designs and thus can be skipped when necessary.

**Procurement phase:** Procurement for a service should come from an actual need for the service. In the case of public services, the need should come from the citizens. Thus, public engagement is required so that the public can have their needs and requirements heard to direct the service procurement. Public administrations can utilise open forums to help communicating with the public. At the same time, the administrations can have initial tenders to start screening for possible service providers and developers. There are two recommendations for these steps that


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should be followed. The first important aspect is an *overall IT vision* for managing and purchasing digital solutions. This is to avoid buying several software programs that perform the same function. The second important recommendation is to have a *working group* that supports the ongoing projects in public administration. The working group would be responsible for digital inclusion, involving target users, meeting their needs, and analysing and benchmarking the digital services with the help of various tools.

There is one best practice during the procurement phase, which is to *choose the service provider carefully*. With public services, the development process is long, and the public administration will be contractually tied to the chosen service provider; thus, the selection process should be done carefully and not just based on the cheapest offer. The service provider should not have a negative public image, as that would negatively affect the user's acceptance and adoption of the service.

*Main challenges* during the procurement phase are the *standards and regulations, technical competency and available technology* and *age difference.* Different locations have their own regulations on public service procurement and different digital services may have their own standards that should be followed, especially if they connect to other existing services. While these are minor challenges, they should be carefully considered by the public administration before starting the overall process. On the other hand, the available technology will be a major barrier for the procurement phase as the cost and feasibility of providing a desired solution will depend on what technologies are possible to use. Similarly, the age difference of the citizens should be considered at this phase. If most users are different ages from young to old, the service may need to be provided utilizing new and old technologies to accommodate the different age groups.

**Design phase**: The second phase in the service development is the design phase. The most important aspect of the design is to engage the public and involve them in the design process as much as possible. If public opinion is not listened to, the design will eventually have problems as the stakeholders and different scenarios have not been thoroughly considered. It is highly advised that the technical partners would develop user stories based on the public's feedback and opinions. This will boil down to the best practice mentioned: *do not forget the human elements in the process,* and the design team should have a user-centric approach.

There are three recommendations for this phase :

- Solutions should be cheap for users and service providers.
- There needs to be value for users.
- The payment should be included in the taxation if it is a paid service.

All recommendations somehow tie to money and value. Users are incredibly hesitant to pay for services that are free from other sources, which is why the payment should be "invisible" to the citizens. In the case of a private service, there is some leniency towards having to pay, but even then, the cost should be lower than that of competitors. At the same time, the service should also be cheap for the service providers as the costs should be kept at a low level in public services. Finally, there needs to be some tangible or intangible value provided to users that would make the new service more appealing compared to old or existing services. The value can be as direct as *saving money* or more indirect, such as more accessible *and faster to use*. If there is no value or it is unclear for the users, they are less likely to use and adopt the service.

*Main challenges* for the design phase are *level of knowledge and education*, *technical competency and available technology*, *value vs. required work* and *previous experience*. During the design phase, the user needs should be considered as much as possible, thus the main barriers are all user related, which is why the best practice is to not forget the human elements.

**Development phase**: The third phase is finally the development phase. As with design, listening to the public's voices and opinions is essential. The technical partners must follow the


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legal and ethical standards within the development process that apply, considering the use case and nation(s) involved. For example, medical standards and regulations must be followed when dealing with medical data. There are no recommendations for development as the ethical and legal standards are mandatory. However, two best practices should be followed. Firstly, the development process should *be transparent* for users. This is extremely important in public services as citizens have a right to know how public service development is progressing. This is not so vital with a private service, but it is still a good thing to do. Transparency refers to how the service is developed, how it would work, and how the end-user's feedback has been considered. With transparency, it is easier to get feedback and tackle possible negative thoughts and concerns from end-users before they become an issue.

The other best practice is *testing during development.* Testing should always be done during any kind of development. The testing does not have to be external, but it is advisable to have some end-user members participate. Catching problems and issues and fixing them at this phase is much cheaper than fixing them after the service has been launched.

*Main challenges* during development phase are *standards and regulations* and *privacy and security concerns*. There are different national and international regulations and standards, such as GDPR and ISO standards, that may be necessary to be followed during the development. Similarly, possible privacy and security measures should be developed to alleviate any concerns users may have.

**Instantiation phase**: Instantiation is the process of initialising software with the initialisation arguments, which is the digital service provided in this case. During the instantiation, the designed service is put into the platform, which is then used by the end-users after launch. The main recommendation for this phase is to *have a backup platform or solution* available before the next testing phase (end-user testing) in case there are some problems with the instantiation or testing of the service.

*Main challenge* during the instantiation phase is the *available technology.* When launching a service that is connected to existing systems, the connectivity between the systems can hinder the actual instantiation. Another challenge that may appear is that the designed service utilizes software that is not available on the instantiation site, such as a different operating system.

**Testing phase**: The second last phase refers to the external testing phase. During a service development cycle, there may be multiple design-development-testing iterations (agile development), which this blueprint does not separate. So, while there is only one testing phase in this blueprint, multiple testing phases may occur during the overall development. With external testing, the most critical factor is *involving the public* with a diverse representation group as much as possible. These testers can later *become ambassadors* and promote the service to other users. Public administrations should invite as many citizens as possible to participate, and technical partners can gather user interactions and feedback during the external testing to improve the service further. For the testing phase, there are no specific *main challenges* but rather the challenges mentioned in the previous steps are still in effect.

Figure 5 presents the second half of the blueprint, which shows the final implementation, launch, and service usage phases.

**Implementation and launch phase**: There are no specific actions for the actual implementation and launch, as these depend highly on the type of service being launched. However, there are best practices that should be followed: *consistency in the project launch schedule and technical/informatics consistency*. The project launch schedule means that the services should be launched consistently apart from each other to avoid overwhelming the users with new services coming up erratically after a week or month. On the other hand, technical and informatics


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consistency refers to the consistency with which the services are documented, and they should follow a consistent scheme. At this phase, the *main challenges* from previous steps should have been resolved.

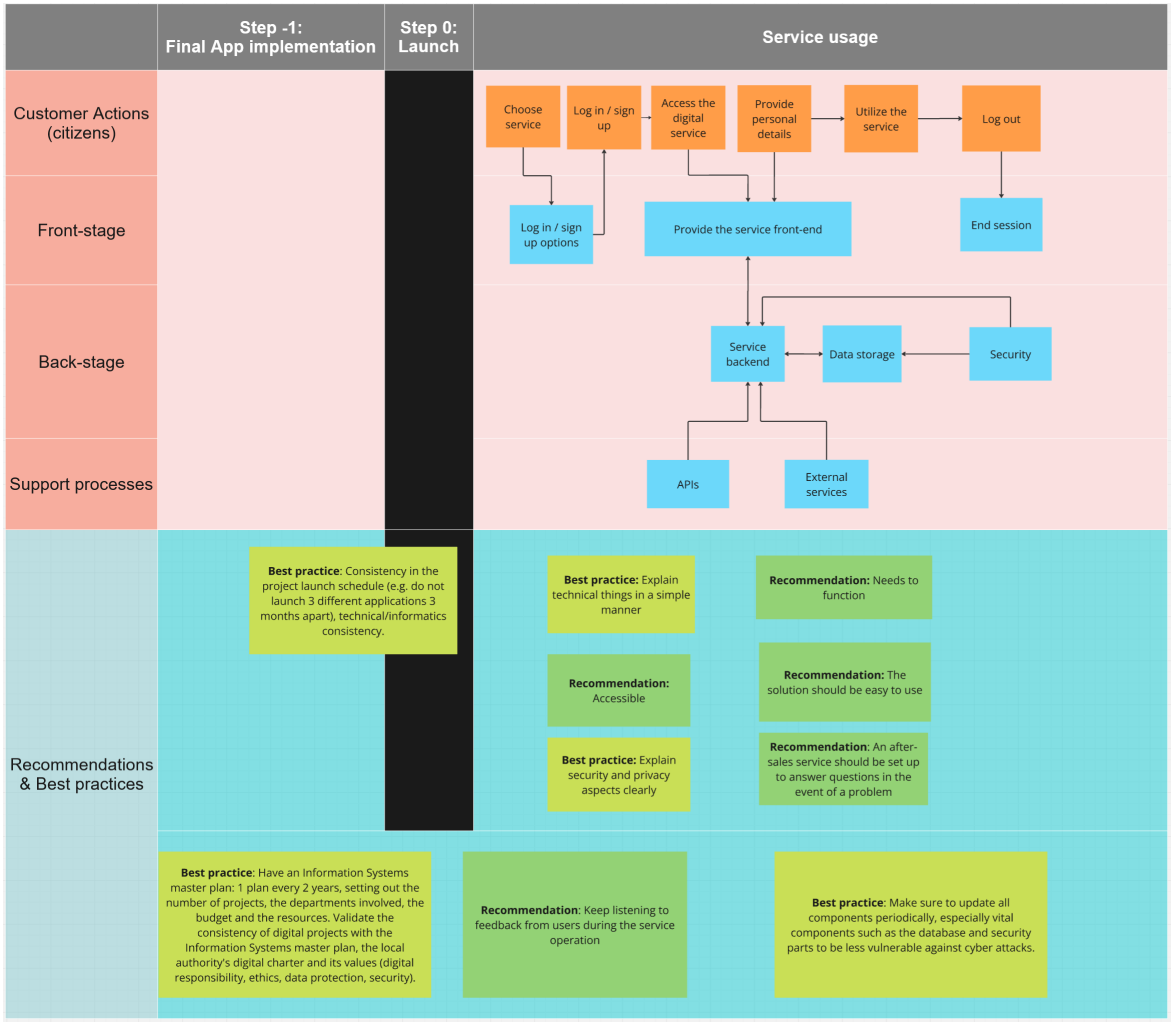



**Figure 5.** Service blueprint, part 2 after final.

**Service usage**: The service usage phase is a generalised process blueprint and should appropriately be adapted to the specific case and context. However, the best practices and recommendations apply regardless of the context. There are four recommendations to be followed and two best practices:

- Needs to function
- Accessible
- Easy to use
- Service to answer questions in the event of a problem
- Explain technical things in a simple manner
- Explain security and privacy aspects clearly

First, it is imperative that the service functions or that no user is willing to adopt it. The functioning does not only mean that the service needs to be online but also to provide the service it is meant for. If these are not met, the service will not be used. Another important aspect is that the service is *accessible*. Many citizens have different constraints, such as not having specific


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technological devices, having trouble differentiating colours, and language barriers, among others. All these must be resolved, and most citizens must be catered to. Thirdly, the service should be *easy to use*, which will significantly impact user acceptance. Accessibility overlaps with ease of use, but there are also differentiating actors, such as the service providing information on what the user is supposed to do next and simplifying the service process as much as possible. The final recommendation is that there should be a *service to answer questions in case of a problem*. Many users will encounter some issues with digital services. There needs to be a way to receive help from the service provider, such as directly contacting someone or having some artificial intelligence assistance.

The two best practices relate to how users differ in knowledge and competency. Most users find privacy and security paramount, but everyone will have different knowledge and understanding. Hence, it is up to the service provider to *explain security and privacy aspects clearly,* not with just a lengthy privacy policy written in legal jargon that no one can read. Aside from the official privacy policy documentation available for all users, there should be some simplified explanation. Similarly, *technical things should be explained simply*, especially when the service uses some new technology, such as artificial intelligence, that some people would not comprehend properly and might have some wrong preconceptions.

*Main challenges* during the service usage phase are *level of knowledge and education, value vs. required work* and *previous experience* assuming the challenges of previous steps have been resolved. The overall service should provide value for the user, and the value should outweigh the work required to be able to utilize the service. For example, if the steps to start using the service are too complicated, users will be less likely to onboard with the service. Similarly, the previous experiences users have had with similar services will positively or negatively affect their willingness to use the service. Even if the public service is vital, users will harbour negative feelings towards the service if the challenges have not been resolved. In addition, the service should consider the different knowledge levels of the users and provide explanation on how the service works.

**General best practices and recommendations:** In Figures 3, 4 and 5, there are combined six general best practices and recommendations in the final row that apply to all phases:

- Have an overall IT vision to centralise software purchases and management
- Creation of a digital working group that supports digital projects
- Communicate with users throughout the project with various channels, such as social media, websites, and events.
- Have an information systems master plan every two years that sets the number of projects, involved departments, budgets and resources.
- Keep listening to feedback from users even when the service is operational
- Ensure that all digital components are updated periodically, especially vital components such as database and security.

## 5.2 Recommendations for use, limitations and other considerations

The proposed service blueprint acts as a template for a digital service design and is meant to be used as a supporting tool for service development. The blueprint is generic for digital public services and can be easily adapted to specific use-case scenarios as long as the necessary steps for the process have been completed. The blueprint can be used as a comparison or reflection point to check if something has been forgotten or could be changed in the process. While the genericity of the blueprint is a benefit as it is easy to adapt, it can also be a drawback as it is not an exact fit for any digital public service and cannot be used as-is. Although the blueprint has been designed with public services in mind, it can also be used for private services if the intention


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is to have end-users (the public) participate in the design and development phases. The following questions are meant to give guidance to the most important challenges for each phase in the service design and development:

- Procurement phase: Have you examined what technologies and providers are available and what standards and regulations you have to follow?
- Design phase: Have you considered the different types of users that will be using your service?
- Development phase: Have you considered the privacy and security measures that will be implemented?
- Instantiation phase: Have you made sure the available and existing technology is compatible with the designed service?
- Testing phase: Have you included a suitable representation of the whole end-user group in the testing?
- Implementation and launch phase: Have all the design and development challenges been resolved?
- Service usage phase: Are you providing enough value for users with less work needed, especially compared to existing services?

Other tools are recommended to be used with the service blueprint to help with service development. For example, performing stakeholder analysis and adequately framing the problem or need with the help of W6H questions can support service development. Another relevant part is developing plans to involve the end-users in the process and adequately designing the collaboration methodology.

As limitations, the blueprint is not an all-inclusive service design tool that guarantees success. The main goal of the proposed blueprint is to increase the acceptance and adoption of the desired public service through the provided recommendations and best practices, and these can be different based on the specific context and should be separately viewed for each scenario. There are also no specific tools and methods included in the service blueprint, which depend on the service designer and provider. For example, design and development processes will vary across organisations, and the service blueprint does not consider any specific process that should be followed. Similarly, the stakeholder and needs analyses can be done using any tool or process the designers and developers are accustomed to. Finally, while the proposed blueprint is based on data collected during public service development in the IMPULSE project, the data is limited to that specific project and its case locations. The proposed blueprint is also not tested and verified on if following it will improve the acceptance and adoption when developing a public service.

## 6 Conclusion

This paper explored the acceptance and adoption barriers and challenges in digital public services, based on the findings in the IMPULSE project, and proposes the usage of a service blueprint to help improve the acceptance and adoption rates. There are multiple factors and variables that will create barriers and challenges for digital public service adoption. Different demographics, levels of knowledge, concerns, and needs significantly impact how likely users are to adopt the service. Using TAM, it is possible to evaluate the user's likelihood to use a service and how these given factors connect to the overall user acceptance and adoption of technologies and the service itself.

After identifying eight factors during the IMPULSE project, different recommendations and best practices were extracted to overcome the barriers and challenges. These recommendations and best practices are tied to user experience, user needs and requirements, and communication.


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When correctly following these, the users would be more likely to accept and adopt the digital service. With the help of the service blueprint technique, a graphical representation of a general digital public service design was developed to with identifying possible problem areas within specific phases during the service design and development process. If the challenges can be identified before the development process goes too far, the easier and cheaper it will be to make changes.

This paper proposes a modified service blueprint model for digital public services to help increase the acceptance and adoption rates of digital public services. The model is simplified to be as generic as possible to be adaptable for any given case. The model includes the different phases of service design and development starting from procurement to the service usage, and different steps that the different actors may take. In addition, the model integrates the gathered recommendations and best practices to the different phases in the service blueprint design helps to keep track of what should be done to meet the user needs and requirements and not have to deal with the challenges and barriers that would otherwise come up.

This paper is meant to help with future service design and development of digital public services. Although the blueprint is designed for public services, it is possible to adapt to the private sector. Adapting the service blueprint for hybrid services that use physical and digital steps is also possible. It is also possible to utilize the finding of this paper without the use of the service blueprint.

As limitations, the data collected is only from one project that utilized eID service. The blueprint itself is a conceptual model and not verified on if following it will increase adoption or acceptance in the end or if it will help the developers during the design and development phase. Additionally, the blueprint acts more as a template future use and needs to be adapted to each specific context. For future work, the effectiveness of the designed blueprint should be tested and verified. Additionally, the different challenges and barriers should be verified to find out if they change or there are more based on the different use-cases.

### AI Disclosure Statement

The authors acknowledge the use of Grammarly for language refinement. The authors confirm that all intellectual content, interpretation, and conclusions are their own, and that AI use did not compromise the originality, accuracy, or scientific integrity of the work. The author(s) take full responsibility for the content of the research work.

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## Biographies



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